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MILITARY SPECIFICATION

CRANE, RAILWAY, DIESEL-ELECTRIC, 40-TON CAPACITY (56-1/2-INCH GAGE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a diesel-engine-driven, self-propelled, standard gage railway crane, with a hoisting capacity of 40 tons, that is capable of conversion to pile driving service.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

W-B-131 - Battery, Storage, Vehicular, Ignition, Lighting and Starting.
RR-W-410 - Wire Rope and Strand.

MILITARY

MIL-V-173 - Varnish, Moisture- and Fungus-Resistant (For the Treatment of Communications, Electronic, and Associated Electrical Equipment).
MIL-H-21103 - Hammer, Piledriver, Self-Powered, Diesel.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 2230

DISTRIBUTION STATEMENT A.
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STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Service.
FED-STD-595 - Color.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Railroad Administration (FRA).
United States Safety-Appliance Standards.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as Department of Defense (DoD) adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B15.1 - Safety Guards for Mechanical Power Transmission Apparatus.
B30.5 - Crawler, Locomotive and Truck Cranes.

(Application for copies should be addresssed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

ASSOCIATION OF AMERICAN RAILROADS (AAR)

Manual of Standards and Recommended Practices.

(Application for copies should be addressed to the Association of American Railroads, 59 E. Van Buren Street, Chicago, IL 60606.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A275 - Magnetic Particle Examination of Steel Forgings.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

MG 1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The railway crane shall consist of:

a. A railway car body, including underframe and steel plate deck; equipped with two 4-wheel swivel trucks for travel on a standard gage railroad track, and the necessary draft and brake gear.
b. A full revolving superstructure with diesel engine powerplant and crane equipment mounted thereon.

The crane shall be capable of yard switching service and train travel on mainline tracks. When operated singly or for switching service, the crane shall be driven by electric motors, one mounted on each truck and controlled from the operator's cab, powered by an electric generator mounted on the revolving superstructure and driven by the diesel engine. The operations of hoisting, swinging, booming, and propulsion shall be independent of each other and be capable of being performed simultaneously, subject only to the speed

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limitations. The crane shall be designed and constructed so that it can be converted to pile driving service. The overall dimensions of the crane shall be the minimum consistent with the required ruggedness and serviceability. The overall length, without boom, shall not exceed 30 feet. The radius of the cab end swing shall not exceed 13 feet. The crane shall be capable of operation, without malfunctioning or damage to any component or assemblies at ambient temperatures from -20 degrees Fahrenheit (°F) to +125°F and shall be suitable for extended storage in any ambient temperature between the limits of -65°F to +160°F. The crane shall comply with applicable AAR Manual of Standards and Recommended Practices and the DoT United States Safety Appliance Standards. Each crane shall be complete and ready for operation when additional ballast is installed at the operating site.

3.2 Standard commercial product. The railway crane shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the railway crane being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3 First article. When specified (see 6.2), the contractor shall furnish one railway crane for first article inspection and approval (see 4.2.1 and 6.4).

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Safety. All rotating and reciprocating parts and all arts subject to high operational temperatures, that are of such nature or are so located as to be or become a hazard to ther operating or attending personnel, shall be adequately guarded or insulated in accordance with ANSI B15.1. Ladders, steps, and handholds shall be provided for easy access to the control station. All surfaces to be walked on shall be an antiskid surface. A weatherproof electric or air operated horn shall be provided for signaling and warning

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purposes. The control shall be conveniently located within reach of the crane operator. All safety features shall be in accordance with applicable DoT, OSHA, AAR, and ANSI B30.5 standards.

3.7 Threads. Screw-threads for all thread securing or attaching devices and threaded parts shall be in accordance with FED-STD-H28. Machine threads shall be the standard national coarse or fine thread series. Pipe threads shall be the standard national taper pipe thread series.

3.8 Capacities and functions. The crane shall have hoist capacities, when used with 50-foot boom, as specified in table I. The minimum radius of the hoist shall be not more than 12 feet. The maximum radius shall be not less than 50 feet (see 3.15). The radii shall be the horizontal distance from the center of gravity of the suspended load to the axis of rotation. The hoist hooks of the crane shall be capable of working at any radius between the minimum and maximum to an elevation of 25 feet below the top of the rail. The maximum unit stress imposed by rated loads shall be in conformance with the requirements of SAE J987. The crane shall be capable of hoisting, traveling, and luffing separately or simultaneously, as may be desired, when carrying any load beyond the 20-foot radius without outriggers and within the capacities and radii specified. The center of gravity of the fully ballasted crane shall be not more than 22 inches from the center of the track with the boom at minimum radius at right angles to the track, with no load on the hook. The crane shall be capable of operating in either direction over track curves of 60-foot radius, and shall be capable of traveling up a straight grade of 5 percent with no load on the hook. Maximum travel speed of the crane on level track shall be not less than 10 miles per hour. Tipping load capacity of the crane shall be not less than 118 percent of the hoist capacity in the least stable direction without outriggers and 125 percent with outriggers.

TABLE I. Hoist capacity.

Radius (feet)	Lifting capacities (lbs) with outriggers	Without outriggers
12	80,000	60,000
15	67,000	44,000
20	51,000	31,000
25	42,500	24,000
30	37,500	19,000
35	30,500	16,000
40	24,500	13,000
45	19,500	10,500
50	18,500	9,000

3.9 Car body. The design and construction of the car body shall conform to the AAR Manual of Standards and Recommended Practices except as otherwise specified herein. Where bolted connections are used, keys, lockwashers, or double nuts shall be used. On draft gear and brake systems, locknuts shall be provided over the single nuts.

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3.9.1 Underframing. The underframing shall consist of longitudinal and transverse steel beams welded together and plated so as to form a substantial structure for supporting the roller path, center pin steel casting, where used, and counterweight receptacles.

3.9.2 Draft gears and couplers. The draft gears shall be of the friction type with cast steel yokes. Coupler shall conform to the AAR standard with 6 1/4-inch by 8-inch shank and type E head. The coupler operating devices shall be suitable for use with the couplers provided and shall operate from both sides of each end of the car. Couplers which are not of the safety flange type shall be provided with guards.

3.9.3 Outriggers. Each car body shall be fitted with four outriggers, two on each side of the car. The outriggers shall be I-beams or box type beams arranged to slide under the car when not in use. Outriggers shall clear the top of the rail by not less than 10 1/2 inches. Four sliding wedges, adjustable jackscrews, or cams shall be provided for stabilizing the car when making lifts. Outriggers shall be operable without the need for the operator or assistant to get under the car body to set up lifts. When specified (see 6.2), outriggers shall be hydraulically controlled. Beams shall retract and extend from under the car body by hydraulic power. Hydraulic stabilizer cylinders shall be provided for each outrigger beam.

3.9.4 Trucks. Trucks shall be of the 4-wheel side frame type for 56 1/2-inch gage track, with cast steel bolsters and frames, 33-inch diameter rim-toughened solid wrought steel wheels, 6- by 11-inch journals, and forged carbon steel axles, and shall be in accordance with AAR specification M-107. Center bearings shall be integral with the bolsters and shall be provided with accessible means for lubrication. Bolster side bearings or rubbing blocks shall be provided.

3.9.5 Travel brakes. The crane shall be fitted with straight independent airbrakes employing a self-lapping type 3-way directional control air valve (hand operated). Brakes shall operate on all wheels, shall be complete with all necessary fittings, drain line, air pipe, and hose. Handbrakes and ratchets shall operate on all wheels of one truck. Brakeshoes shall be cast iron steel reinforced backs. Straight air brakes shall be controlled from the operator's position in the cab. Braking force, with 59 pounds per square inch gage in the brake cylinder, shall provide not less than 13 percent minimum of the gross weight of the crane. When specified (see 6.2), the crane shall also be provided with BBG class II (automatic) airbrakes.

3.9.6 Turntable. The manufacturer's standard commercial turntable shall be provided.

3.10 Rotating platform. The rotating platform, upon which the mechanisms and prime mover above the car body are mounted, shall be a heavy 1-piece casting or heavy weldment. A suitable load roller and hook roller arrangement shall center the rotating platform and transmit loads under all rated conditions of loading to the car body roller path. The rotating platform shall be of ample size to accommodate conveniently, without crowding, the machinery, counterweight, levers, and other equipment necessary for the proper

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operation of the crane. The floor of the rotating platform adjoining the cast steel base shall be of sufficient strength to support the counterweight and auxilliary machinery, and shall be connected securely to the cast steel base.

3.10.1 Counterweight and ballast. Any necessary counterweight shall be solid cast iron. The contractor shall furnish and install all necessary counterweights. If additional counterweight is needed for maximum radius lifts, waterweight space shall be provided in or on the rotating superstructure. If ballast is necessary, waterweight spaces for ballast shall be provided within the car body. The size of spaces for ballast and additional counterweight shall be based on a material weight of not more than 200 pounds (lbs) per cubic foot.

3.10.2 Cab. The cab shall provide weather protection for the operator and the lifting machinery. The cab shall have sufficient windows to provide at least 120° horizontal and 90° vertical vision for the operator. The ballast shall be of the safety type and shall be held in place by rubber channels or other suitable means for weather proofing and for easy replacement. A powered windshield wiper shall be installed with the control within reach of the operator. Cab doors, whether of the sliding or swinging type, shall be adequately restrained from accidentally opening or closing when the crane is traveling or operating. An adjustable and comfortable seat shall be provided for the operator and shall be located for maximum unobstructed visibility. The cab shall be furnished with the manufacturer's standard heater and defroster. Cab shall meet all applicable ANSI and FRA requirements.

3.10.3 Controls. All motions of the crane, except propulsion, shall be controlled by compressed air or full hydraulics. For air system, brakes on the hoists shall be either mechanical or air set, with mechanical emergency brake operating on the same pedal. The normal operating pedal pressure shall be proportional to the braking effect and not less than 20 lbs for full braking. Control valves shall be of the heavy-duty compensating type for air systems and self-centering for hydraulic type systems. Air systems shall be complete with all necessary piping, flexible hose connections, compressors, reservoirs, pressure reducers, drain valves, quick-release valves, unloaders, pressure gages, moisture traps, and strainers.

3.11 Lighting system. The lighting system for the crane shall be the manufacturer's standard which meets current applicable safety regulations and consists of at last 2 each 150-watt floodlights mounted on the crane cab, 2 each 150-watt floodlights mounted on the crane booms, 2 each cab interior lights and the manufacturer's standard, commercial railway approved running lights. The lighting system shall be furnished with not less than 1800-watt alternator.

3.12 Boom angle indicator. The boom angle indicator shall meet the requirements of SAE J375. The indicator shall display the angle of the boom relative to the horizontal plane and shall be clearly visible to the operator. A chart located in the cab near the operator shall be provided showing rated lifting capacity at various angles for various boom lengths with and without outriggers.

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3.13 Machinery. Machinery shall be capable of operating the crane at the specified speeds and at the maximum specified loads, with ease, safety, minimum noise, and vibration. All parts shall be designed so that they may be readily disassembled, assembled, adjusted, and repaired. Also parts shall be accessible for inspection, cleaning, and lubrication. All fastenings and parts which are likely to become loosened from vibration shall be secured by devices which will insure the safety and reliability of the feature concerned. All frames carrying power shafts shall be of cast or welded steel. Holes for bearings shall be bored to micrometer sizes and details shall be made to gages which will insure duplication of parts when needed for repairs. If chain drives are used, a means for chain adjustment shall be provided and chains shall be enclosed with a removable metal cover.

3.13.1 Hoisting mechanism. The crane shall be provided wth two independent, free running, hoisting drums mounted on the same shaft, with antifriction bearings, each driven through a separate friction clutch and provided with a separate brake. The main hoist mechanim shall have a minimum hoisting capacity of 40 tons. The speed reduction of the gearing, drum diameter, and the rope reeving shall be such as to provide a hoisting speed for the main hook, when lifting the maximum load, of not less than 30-feet per minute when six part rope reeving is provided. The load line shall be dead ended at the boom point in a becket (rope socket). The main load line shall be 4-part or 6-part reeved.

3.13.2 Power load lowering. The primary load hoist drum shall be equipped with a powered load lowering drive and suitable controls to provide precise controlled power load lowering of loads on the main hook. The powered load lowering mechanism shall be controlled from the operator's station and shall be capable of lowering all rated loads with the manufacturer's recommended reeving. The powered load lowering mechanism shall be capable of lowering loads at a smooth, steady controlled, rate of speed without jerking, bouncing, or slipping the load.

3.13.3 Boom hoisting mechanism. The boom hoisting mechanism, brake, and lock shall conform to the requirements of ANSI B30.5. The mechanism shall consist of friction clutches, gearing, drum, brake, and ropes. The boom hoisting drum shall be driven by hardened gears which shall be fully enclosed in an oiltight case. The shaft shall be bearing mounted. The boom hoisting mechanism shall be designed so that the maximum radius can be hoisted to the minimum radius in not over 2 minutes when equipped with a 50-foot boom.

3.13.4 Rotating mechanism. The rotating mechanism shall be operated through gearing and friction clutches and shall be capable of rotating the crane continuously in either direction. Brakes or locks whereby the rotating mechanism may be locked shall be provided. The rotating pinions shall be integral with, or splined to, the vertical rotating shaft. The strength of the mechanism shall be such as to hold the boom at the maximum radius against 15 pound-force per square foot wind pressure. The crane shall be capable of making not less than 2 1/2 revolutions per minute when the engine is operating at its rated speed.

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3.13.5 Travel mechanism. The crane shall be equipped with a variable speed mechanism controlled from the operator's platform. The mechanisms shall be arranged so that the change in speeds can be made easily while the crane is in motion. Each travel mechanism shall drive two wheels in either direction on each truck through suitable gearing. Travel gears shall be supported in a cast steel housing. The crane shall have a minimum starting drawbar pull on a straight, level, dry track of 12,000 lbs.

3.13.6 Transmission. The crane electrical transmission for propulsion shall be of approved type and design for transmitting of engine power to the wheels for travel in either direction. Transmission shall consist of a main generator directly connected to the engine, and connected through the control apparatus, to traction motors mounted on the trucks and gears to the drive axles. Electrical rotating equipment shall comply with AAR Manual of Standards and Recommended Practices.

3.13.7 Bearings. Bearings for the mechanisms shall be provided with means for pressure lubrication and shall be designed to prevent the dripping of oil or grease. Bronze and antifriction bearings shall not be used in combination where the wear of one type would adversely affect the other.

3.13.8 Gears. Gears shall be of steel, except the worm gears which shall be of bronze. All gears shall be cut, except the rotating gear and pinion which may have cast teeth. Spur and bevel pinions and power shaft bevel gears shall be of steel forgings or heat treated alloy castings.

3.13.9 Wire rope. Wire rope shall be 6 x 19 class, improved plow steel, independent wire rope, uncoatd. Design factors, with manufacturer's recommended reeving shall comply with ANSI B30.5. The rope length on both hoist drums shall be sufficient to lower the hook 25 feet below the top of the rail with the boom at minimum radius and the hook block reeved with the maximum parts of line.

3.13.10 Hook blocks. The main hook block shall have adequate weight to insure payout of the cable without load on the hook when the hook is in any position. The hook shall be forged steel swivel type, designed to handle at least four times the maximum safe load based on ultimate tensile strength and shall include a safety latch on the hook. The main hook block furnished shall permit disassembly for visual and nondestructive testing. A whip line swivel hook and overhaul weight shall also be provided with rated capacity equal to the single line load of either drum. Hooks shall not be painted.

3.13.11 Clutches. Clutches shall be capable of holding not less than 125 percent of the maximum load without slippage, shall be arranged for easy adjustment, shall have a quick and positive release, and shall not drag or seize. The clutches shall be arranged for renewal of wearing surfaces.

3.13.12 Brakes. Brakes for hoisting shall be operatd by foot pedals with ratchets. The rotating brake shall be operated by either a pedal or hand lever. Ratchets shall be provided if brakes are air set and mechanical means are to be used in emergency operations. Main and auxilliary brakes shall be designed to hold a 50 percent overload. Closures shal be provided over brakes as necessary. The boom hoist brake shall set automatically.

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3.13.13. Operating levers. The air system shall be used for controlling the hoisting, rotating, and luffing mechanisms. Variable voltage control for propulsion shall be provided. Level control systems, if used for other purposes, shall be so designed that levers may be thrown with the minimum effort and minimum travel as specified in ANSI B30.5. Control mechanisms shall be grouped on the operator's platform for operation from one position in conformance with SAE J898.

3.14 Engine and compressors. Engine and compressors, when furnished, shall be arranged to be accessible for repairs and operation. Gages shall be located so that they can be seen easily from the engine starting position or readily visible to the operator. Engine exhaust gases shall be piped to the outside of the cab and discharged in a direction away from the operator. All exhaust pipes shall be guarded or insulated in areas where contact by personnel is possible in the performance of normal duties. Exhaust pipes shall be fitted with rain caps. Hot manifolds shall be covered.

3.14.1 Diesel engine. The crane shall be diesel-engine-powered. The power and speed required of the engine to meet the performance requirements of the crane shall not exceed the horsepower (hp) values at which the engine has been rated. The diesel engine shall start in any temperature above -20°F. Starting aids shall be either electric glow plug or ether primer. When an ether priming system is used, it shall be of the measured shot type with storage capacity of not less than 16 fluid ounces. The engine shall be furnished with the manufacturer's standard accessories and, as a minimum, complete with the following accessories:

a. An instrument and control panel located at the operator's station fitted with, but not limited to, the following:

(1) Start stop control.
(2) Liquid quantity fuel indicator.
(3) Lubricating oil pressure indicator.
(4) Cooling liquid temperature indicator.
(5) Battery charging indicator.
(6) Tachometer.
(7) Hour meter. An engine hour meter which has a totalizing mechanism of not less than 9,999 hours shall be furnished to register accurately the number of hours of engine operating time. Meter shall be of strong rugged construction to insure continuous trouble-free performance under the most extreme equipment operating conditions indicated in this specification.
(8) Safety control devices of the visual or audible type for high coolant temperature and low lubricating oil pressure.

b. A complete 12- or 24-volt electrical cranking system with heavy-duty battery(s). When specified (see 6.2), battery(s) furnished shall be dry charged in accordance with W-B-131 and the filler caps sealed to prevent the intrusion of atmospheric moisture.

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c. A battery charging alternator with a rating of not less than 35 amperes.
d. A 2-stage, dry-type, intake air cleaner with service indicator.
e. A raincap is to be furnished on vertical exhaust systems.

Fuel tank capacity sufficient for not less than 10-hours normal operation shall be furnished.

3.14.2 Travel motors. The travel motors shall be of the railway traction type, designed specifically for the service, and shall be series wound, class H insulation, open protected, self-ventilated, and have a commutation field. The 1-hour rating shall be not less than 110 hp. The temperature rise when under full load shall not exceed the ambient temperature by more than 140° Celsius. The travel motors shall be flange mounted to the transmission. The minimum distance between the top of rails and travel motors and gear cases shall be not less than 4 inches.

3.14.3 Rating. The rating of the alternator and motors shall be determined in accordance with NEMA MG 1.

3.14.4 Travel control. The crane travel control shall have not less than four speed points, and shall be operated by means of standard control handles and buttons from the crane cab. The electrical control switches, contactors, conductors, and associated switchboard components shall be adequate for the loads required.

3.14.5 Electrical wiring. Wiring and equipment installation shall conform to the applicable recommendations of the NFPA 70. Insulation shall be oil resistant. Collector rings for transmitting power to the motors shall be located beneath the car body, be heavy-duty type, be readily accessible, and be enclosed in a dust-tight covering.

3.14.6 Air compressor for brakes and controls. For air controlled actuators an air compressor with a capacity of not less than 25 cubic feet of air per minute, driven from the diesel engine, with storage tank, automatic unloader, operator's control valve, safety valve, automatic water drain valve, moisture separator, pressure gage, and all other necessary valves, pipe, and fittings shall be provided on the crane to supply compressed air for the aibrakes and controls.

3.15 Crane boom. The crane boom shall be 50 feet overall in length and constructed in sections. Boom point sheaves shall be mounted on antifriction bearings. The boom shall be constructed to not exceed the maximum stress levels measured in accordance with SAE J987.

3.15.1 Boom extension at tip. When specified (see 6.2), the boom tip shall be provided with a permanent 5-foot extension with a single sheave. The boom tip sheaves shall be provided with cast or forged, flared, well-rounded guards to guide and protect the rope. An auxilliary single line hoist with a 5-ton hook swiveling on an antifriction bearing shall be provided. The rope shall be reeved over the auxilliary drum and the boom extension tip sheave, and shall be suitable for re-reeving for bucket service.

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3.15.2 Center boom extension. When specified (see 6.2), a 20-foot center extension, of the same construction as the boom, shall be provided.

3.15.3 Boom safety stop. A telescopic boom stop shall be furnished and shall be so installed as to limit the maximum boom angle to 85° above the horizontal plane. The complete boom stop with supporting attachments shall be designed with sufficient strength to develop the ultimate strength of the boom in bending at the point of attachment of the boom stop to the boom. The boom stop shall provide a shock-absorbing action and shall remain assembled when the boom point is lowered to the ground. An automatic means to stop boom drum motion, when the maximum permissible boom angle is reached, shall be provided.

3.15.4 Tagline winder. A boom mounted tagline winder with capacity equal to typical crane clamshell bucket capacity shall be furnished.

3.16 Pile driver. When specified (see 6.2), the crane shall be equipped for pile driving serice. The crane shall be capable of operating with a self-powered diesel type pile hammer conforming to MIL-H-21103. The boom tip shall be designed to allow mounting of pile driver lead adapters.

3.17 Tiedown devices. When specified (see 6.2), the unit shall be provided with permanently attached devices to permit tiedown of the unit to the floor or deck of the transportation medium. The tiedown devices shall be of such design, location, and number as to enable the unit to be anchored in such a manner as to prevent shifting or movement in any direction. The devices shall be designed to withstand thrusts of the following magnitude: Foward thrust 8G (G = cargo weight of unit), rearward thrust 8G; upward thrust 2 1/4G, and sideward thrust 1 1/2G. The tiedown devices shall be located as to permit easy attachment of cables or turnbuckles. Tiedown devices shall be indicated by a suitable information plate, by stenciling, or other marking of the unit. The marking shall clearly indicate that the device is intended for tiedown of the unit on the carrier when shipped.

3.18 Lifting attachments. When specified (see 6.2), the equipment shall be provided with lifting attachments to enable the equipment to be lifted in its normal position. The attachments shall be so located that, when hoisted, adequate clearance will exist between lifting slings and all exterior parts of the equipment. When practicable, one attachment of sufficent capacity to carry not less than 500 percent of the total lifted weight shall be provided. When two or more slings are used, each attachment shall be of sufficient capacity to carry at least 250 percent of the weight being lifted. The attachments for multiple slings shall be so designed that the attached slings will converge at a point not to exceed 19 feet above the lowest extremity of the load, when such limitation is possible. Lifting attachments shall be positioned in such direction that the lifting strain will be in line with the longitudinal axis (peripheral plane) of the eye of the attachment. The eye of each lifting attachment shall be not less than 3-inches inside diameter. The equipment members to which padeyes are attached shall be of sufficient strength to withstand stresses in the amount and direction of pull specified for the padeye. Where required, suitable reinforcements shall be used to meet the requirements specified herein. Examples of such reinforcements include but are not limited to the following:

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a. Flat lifting pads shall be bent around the equipment member to give a 90° effect and in such manner as to bear against the equipment surfaces opposite to the direction of the pull. Where it is not practicable to bend the pad around the member, a bolt or bolts of sufficient capacity shall pass through the pad and equipment members. In all cases, the pad shall be welded to the equipment member.
b. Round stock padeyes shall be bent around the equipment member in such a manner as to bear against the equipment surfaces opposite to the direction of pull and shall be welded to the equipment. Where this is not practicable, the round stock shall be threaded and passed through the member and be bolted and welded on both sides of the member.

A diagram showing the lifting attachments and lifting sling shall be inscribed on a copper-base alloy plate securely fastened to the unit. The plate shall indicate the lifting capacity of each attachment and the required length and size of each sling cable marked thereon. A silhouette of the equipment showing the center of gravity shall be provided.

3.19 Toolboxes. One or more toolboxes, as required, shall be provided to carry the tols and accessories required for field maintenance, repair, and operation of the crane. Toolboxes shall be made of steel and have a hinged lid and a latching mechanism to keep the lid secured during operation of the crane. The toolbox shall be securely fastened to the unit in a protected and accessible location.

3.20 Lubrication. Lubrication means shall be provided to all moving parts requiring lubrication. Hydraulic lubrication fittings shall be in accordance with SAE J534. Pressure lubrication fittings shall not be used where normal lubricating pressure may damage grease seals or other parts. The unit shall be lubricated prior to delivery with lubricants designated for use in the 0 to 32°F temperature range. The unit shall be conspicuously tagged to identify the lubricants and their temperature range.

3.21 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.22 Treatment and painting. The crane shall be cleaned, treated, and painted in accordance with good standard commercial practice. The revolving superstructure, operator's cab, and all accessories extending above the car body frame shall be painted No. 13538, Yellow, conforming to FED-STD-595. The car body and crane boom shall be No. 17038, Black. Alternate diagonal yellow and black stripes shall be painted on the boom point, hook block, and counterweight. Paint thickness shall be not less than 2.5 mil.

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3.23 Identification plate. An identification plate will be furnished by the contracting officer for each crane. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix a plate to each crane in a conspicuous place with nonferrous screws, rivets, or bolts not less than 1/8-inch in diameter. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.23.1 Instruction plates/charts. The crane shall be equipped with instruction plates/charts suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws, bolts, or rivets of not less than 1/8-inch diameter.

3.23.2 Crane capacity plate. The crane shall be provided with a crane capacity plate mounted inside the cab in a location visible to the operator. The plate shall be similar to the plate specified in 3.23.1 and shall indicate the safe hook loads of each length at not more than 5-foot intervals for the full range of operating radii. The safe hook loads shall be 85 percent of the tipping loads. Statement of deduction to be made from the safe hook loads for component parts of crane attachments, such as jibs, shall be given.

3.24 Workmanship.

3.24.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.24.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.24.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.24.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

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3.24.5 Machine work. Tolerances for contact and bearing surfaces shall conform to standards prevailing among manufacturers normally producing cranes.

3.24.6 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on each crane when a first article is required (see 3.3 and 6.4). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.4.1, 4.4.2, and 4.4.5, and the packaging inspection of 4.5.

4.3 Examination. Each crane shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

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4.4 Tests. The first article shall receive the tests of 4.4.1 through 4.4.6. Failure to pass any test shall constitute cause for rejection. Tests shall be conducted as outlined in the referenced documents as specified herein.

4.4.1 Limbering-up run. The engine and all operating components shall be given a limbering-up run operation, under no load conditions, for a continuous period of not less than 2 hours. All necessary readjustments shall be made and all minor malfunctions shall be corrected after the limbering-up, in preparation for the following tests.

4.4.2 Test of operational controls. With diesel engine running and air tank fully charged, all controls in the operator's cab for operating the hoist drum, boom and cables, hoist drum brakes, revolving platform, crane travel brakes and speed controls, horn or whistle, floodlights, and engine speed shall be operated at least ten times for determination of proper function. The test shall be made when the crane is completely lubricated.

4.4.3 Lifting capacity. Lifting capacities of the crane shall be verified by subjecting the crane to load stability tests in accordance with SAE J765. The tests shall be conducted in the least stable direction, at a minimum working radius, at the radius equal to the boom length, and three intermediate radii, with and without outriggers. Conformance to 3.8 and table I shall be determined by multiplying the balance point load by 85 percent. Where the lifting capacity is based on structural strength or available hoisting power rather than resistance to overturning, the manufacturer shall submit his test procedure for approval. The procedure shall include all the applicable provisions of SAE J765 and shall be designed to show that the crane will lift at least 118 percent of the rated capacity loads shown in table I without outriggers and 125 percent with outriggers.

4.4.4 Crane hook. All hook components shall be magnetic-particle inspected over the entire area in accordance with ASTM A275. Acceptance standard shall be no defects. A defect is defined as a linear indication revealed by magnetic-particle inspection that is greater than 1/8-inch long whose length is equal to or greater than three times its width.

4.4.5 Travel and brake. The starting drawbar pull and maximum travel speed in forward and reverse shall be measured to determine conformance to 3.8 and 3.13.5. Five complete stops from maximum travel speed in forward and reverse shall be made. Malfunction or excessive temperature rise of the brakes, trucks, wheels, shafts, journals, gearboxes, and bearings or any leakage or malfunction of the brake linkage, air system, or controls shall constitute failure of this test.

4.5 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Commercial. The crane shall be preserved in accordance with the contractor's standard practice in a manner to prevent deterioration and damage. The crane shall be prepared for shipment in a manner which will insure arrival at destination in a satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations.

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6. NOTES

6.1 Intended use. Cranes covered by this specification are intended for wrecking service and heavy lifting. When equipped for pile driving, their use shall be for driving wood piles, steel pipe, and sheet piling.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When first article inspection and approval is required (see 3.3).
c. When hydraulic outriggers are required (see 3.9.3).
d. When BBG class II (automatic) airbrakes are required (see 3.9.5).
e. When battery(s) furnished shall be dry charged or moist charged (see 3.14.1).
f. When boom tip extension is required (see 3.15.1).
g. When a center boom extension is required (see 3.15.2).
h. When the crane will be equipped for pile drive service (see 3.16).
i. When tiedown devices are required (see 3.17).
j. When lifting attachments are required (see 3.18).
k. When treatment for fungus resistance is required (see 3.21).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of DoD Federal Acquisition Regulations (FAR) Supplement 27.410-6, the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DoD FAR 27.410-6 are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one complete railway crane. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:

Army - ME
Navy - YD
Air Force - 84

Preparing Activity:

Navy - YD

(Project 2230-0058)

Review Activity:

Air Force - 99

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-16915D	CRANE, RAILWAY, DIESEL-ELECTRIC, 40-TON CAPACITY (56-1/2-INCH GAGE)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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